

Leopard Gecko (*Eublepharis macularius*)

Difficulty: Easy / Beginner

Leopard geckos are crepuscular, ground-dwelling lizards from semi-arid South Asia. Recent range-wide genetic work indicates that the captive pet-trade lines commonly called *Eublepharis macularius* cluster most closely with wild populations from central and southern Pakistan. Adults are usually about 7-10 inches long and commonly live 15-20 years or longer with good care. In the wild they shelter in burrows, cracks, and other protected retreats, then emerge to hunt and thermoregulate when conditions are favorable.

Modern leopard gecko husbandry has moved beyond the old "10-gallon tank, heat mat, reptile carpet, no light" formula. Research shows that leopard geckos actively thermoregulate, can synthesize vitamin D3 from low levels of UVB, and benefit behaviorally from environmental enrichment. A high-welfare setup therefore provides meaningful floor space, overhead heat, low-level UVB, several secure hides, naturalistic structure, and opportunities to dig, climb, hunt, and choose among different microclimates.

Shopping List

- Front-opening enclosure at least 36" x 18" x 18" for one adult; 36" x 24" x 18" or 48" x 24" x 24" is even better
- Secure, well-ventilated lid or top
- White/clear halogen or incandescent basking lamp
- Dimming thermostat or quality lamp dimmer for heat control
- Low-output linear T5 UVB fixture appropriate for Ferguson Zone 1
- Plug-in light timer; optional 6500K LED for brighter daytime illumination
- Digital probe thermometers/hygrometer and infrared temperature gun
- At least three secure hides: warm, cool, and humid
- Shallow water dish
- 4" or more of a packed semi-arid soil/sand/clay substrate for healthy animals after quarantine
- Slate, cork bark, low branches, rocks, leaf litter, and other stable enrichment
- Varied live feeder insects such as roaches, crickets, black soldier fly larvae, silkworms, and mealworms
- Quality gut-load for feeder insects
- Calcium supplement and a reptile multivitamin containing preformed vitamin A
- Feeding tongs and a digital gram scale for body-weight monitoring

Housing

A single adult leopard gecko should be housed in an enclosure no smaller than 36" x 18" x 18". Larger floor plans such as 36" x 24" x 18" or 48" x 24" x 24" make it easier to provide a broad thermal gradient, multiple hides, and enrichment. Floor area matters most, but leopard geckos will also use low, sturdy climbing features.

A large enclosure does not "overwhelm" a leopard gecko when it is properly furnished. The key is cover. Provide multiple hides and sheltered routes so the gecko can move between warm and cool zones without crossing a completely exposed tank. Heavy rocks and slate should be supported from the enclosure floor or otherwise secured before loose substrate is added, so digging cannot undermine them.

Juveniles can be raised in a full-sized enclosure when food intake and waste can be monitored and the habitat contains abundant cover. Paper-towel quarantine is useful for a new or medically compromised gecko, but a sparse permanent setup should not be considered the welfare goal.

Lighting & UVB

Leopard geckos are crepuscular rather than "lightless" animals. Provide a defined daytime period, typically about 12 hours in winter and up to 14 hours in summer, with complete darkness at night. Daylight and UVB should be concentrated on the warm side, leaving the cool side dimmer and well shaded.

Low-level UVB is strongly recommended. Controlled studies show that leopard geckos synthesize vitamin D3 under UVB even when dietary D3 is available. For normally pigmented animals, target roughly UVI 0.5-1.5 at the basking area. For albino or other hypopigmented morphs, keep basking UVI closer to 0.5-0.7 and provide abundant shade.

Use a low-output linear T5 fixture rather than a high-output desert lamp, and ideally verify the actual UVI with a Solarmeter 6.5. Mesh, distance, reflectors, and lamp age all change exposure. UVB should overlap with the basking zone while at least half of the enclosure remains shaded. Do not use red, blue, purple, or black night bulbs; leopard geckos can see colored light and need darkness for a normal day/night rhythm.

Heating

Use overhead halogen or incandescent heat as the primary daytime heat source. Overhead heat creates a natural surface-to-shelter gradient and allows the gecko to choose between direct warmth and a protected warm hide. A flat piece of slate or stone beneath the lamp works well because it warms during the day and releases heat gradually after the light turns off.

Basking surface: 94-97°F (34-36°C)

Warm hide: 90-92°F (32-33°C)

Cool end: 70-77°F (21-25°C)

Nighttime: a drop into the 60s°F is acceptable for healthy adults; provide lightless supplemental heat only if the room becomes unusually cold

Leopard geckos do not require a heat mat as their sole heat source. Research shows that they actively seek warm refuges during the day and use stored heat later. If a heat mat is used only as backup, control it with a thermostat; it should not replace the overhead daytime gradient. Never use heat rocks.

Humidity

Keep the main enclosure relatively dry and well ventilated, generally around 30-50% relative humidity. Normal short humidity spikes are not a problem when the enclosure dries again, but a chronically damp enclosure can contribute to skin and respiratory problems. Provide fresh water in a shallow bowl at all times.

A humid hide is essential even though the enclosure is dry overall. Place it near the middle of the thermal gradient and maintain roughly 70-80% humidity inside. The lining should be clean, slightly damp, and free of long fibers or loose pieces that the gecko can swallow. Check toes, tail tip, and eyelids after sheds; retained skin in these areas should be addressed promptly rather than allowed to constrict tissue.

Substrate

For a healthy leopard gecko that has completed quarantine, a naturalistic loose substrate can be an excellent welfare choice because it permits digging and supports a more complex environment. A practical mix is approximately 60% pesticide- and fertilizer-free organic topsoil, 30% washed play sand, and 10% excavator clay or bentonite clay by volume. Mix it thoroughly, pack it several inches deep, shape stable terrain while damp, and let it dry before introducing the gecko.

Loose substrate is not appropriate for every situation. Use paper towel temporarily for quarantine, acute illness, or when stool needs close monitoring. Avoid calcium/vitamin sand, ground walnut shell, wood chips, large bark, pure coconut fiber as the only substrate, and reptile carpet. These materials either create ingestion/respiratory concerns, do not resemble the gecko's hard-packed natural ground, or can trap claws and teeth.

Impaction should not be treated as an automatic consequence of all natural substrate. Risk rises when a gecko is sick, dehydrated, kept too cool to digest normally, or offered unsuitable particulate material. Correct temperatures, hydration, feeder size, and substrate choice matter together.

Food

Leopard geckos are insectivores. Feed a varied rotation of live, appropriately sized invertebrates rather than relying on one feeder for life. Good staples and rotation foods include dubia or discoid

roaches, crickets, black soldier fly larvae, silkworms, and mealworms. Hornworms are useful for variety and hydration. Superworms and waxworms are calorie-dense and are better used sparingly, especially in adults that gain weight easily.

Hatchlings and small juveniles: small meals daily or nearly daily, adjusted to growth and body condition

Older juveniles: about every other day or 4-5x/week

Adults: usually 2-3x/week, adjusted to body condition, activity, season, and reproductive status

Offer appropriately sized prey, remove aggressive uneaten insects, and gut-load feeders for 24-48 hours before use. Hunting is useful enrichment. Welfare studies show that leopard geckos use feeding, hiding, and climbing enrichment and prefer enriched naturalistic housing over barren setups; placing smaller prey in multiple locations can encourage hunting without increasing total calories.

Supplements

Feeder insects are typically high in phosphorus and low in calcium, so supplementation is essential. With appropriate UVB, dust most meals lightly with plain calcium or a low-D3 calcium product according to its label. Use a reptile multivitamin less often, commonly every 1-2 weeks for adults and as directed for growing juveniles.

Choose a multivitamin that provides preformed vitamin A (retinol) rather than relying only on beta-carotene. Vitamin A deficiency is an important nutritional problem in insectivorous reptiles, while excessive fat-soluble vitamins can also be harmful, so avoid combining multiple overlapping supplements. If UVB is not provided, dietary D3 becomes more important; however, the preferred approach is measured low-level UVB plus a conservative supplementation program.

Decor

Provide at least three secure hides: one warm, one cool, and one humid. Add cork bark, low sturdy branches, slate ledges, textured rocks, leaf litter, and diggable areas so the enclosure offers more than empty floor space. Make all climbing features stable and low enough that falls are unlikely to cause injury. Cover along the back and sides can also help the gecko feel secure while still allowing observation from the front.

Environmental enrichment supports normal behavior. Controlled studies show leopard geckos use hides and climbing features and prefer enriched naturalistic housing. Safe textures, low climbing opportunities, multiple feeding locations, and occasional small changes to decor encourage exploration.

Handling Tip: Let a new gecko settle and feed consistently before routine handling. Scoop from below, support the whole body, and never restrain or lift by the tail. A dropped tail can regrow, but regeneration uses energy and the new tail will look different.

Cohabitation

House leopard geckos individually for routine pet care. Apparent “cuddling” can actually be competition for a preferred hide or heat source, and subtle competition can cause chronic stress, weight loss, missed meals, injuries, or loss of access to the best microclimate. Males should never be housed together.

Temporary pairing for reproduction is a managed breeding practice, not a general housing recommendation. One gecko per enclosure gives the clearest control over food intake, feces, body condition, temperature choice, and medical monitoring.

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